ICT702 Data Visualisation

Data Visualisation and Design

Module 3

USC

I would like to acknowledge the traditional custodians of the land we are meeting on,

The Gubbi Gubbi/Kabi Kabi peoples,

and pay my respect to Elders past, present and emerging.

USC AUSTRALIA

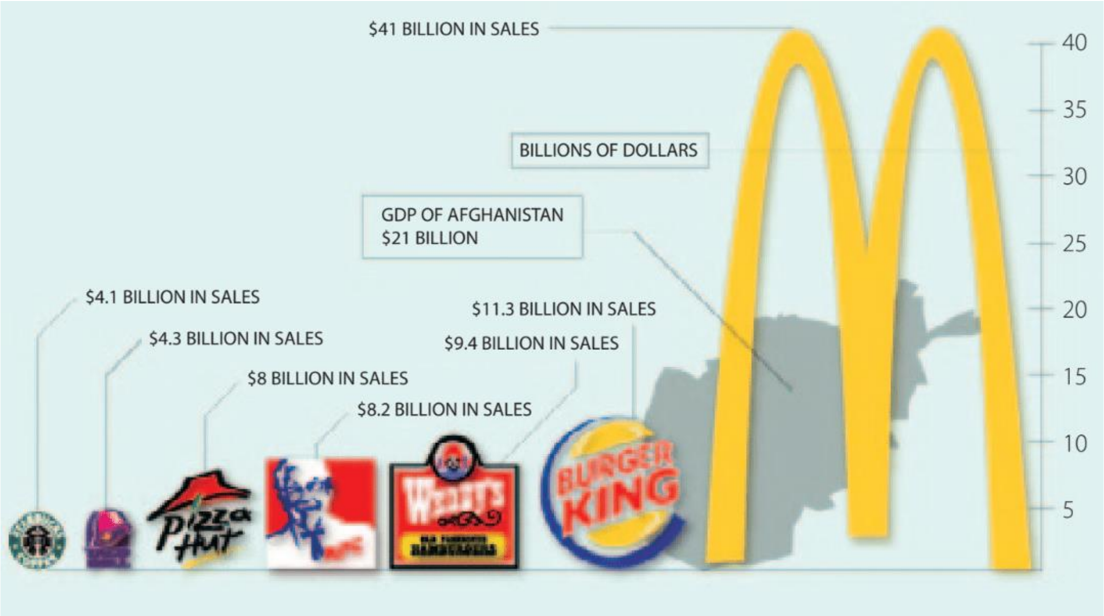
Learning Objectives for Module 3

- Define *preattentive* attributes
- Explain how *preattentive* attributes associated with colour, form, spatial positioning, and movement are used in data visualizations
- Explain how the Gestalt principles of similarity, proximity, enclosure, and connection can be used to create effective data visualizations
- Define *data-ink* ratio
- Explain how increasing the data-ink ratio through decluttering can create data visualizations that are easier to interpret
- Create data visualizations that are easier to interpret by minimizing the required eye travel and applying the concepts of preattentive attributes, Gestalt principles, and decluttering
- Explain why certain types of font are preferred over others for use of text in data visualizations
- List several common mistakes in designing data visualizations and explain how these mistakes can be avoided

Part 1

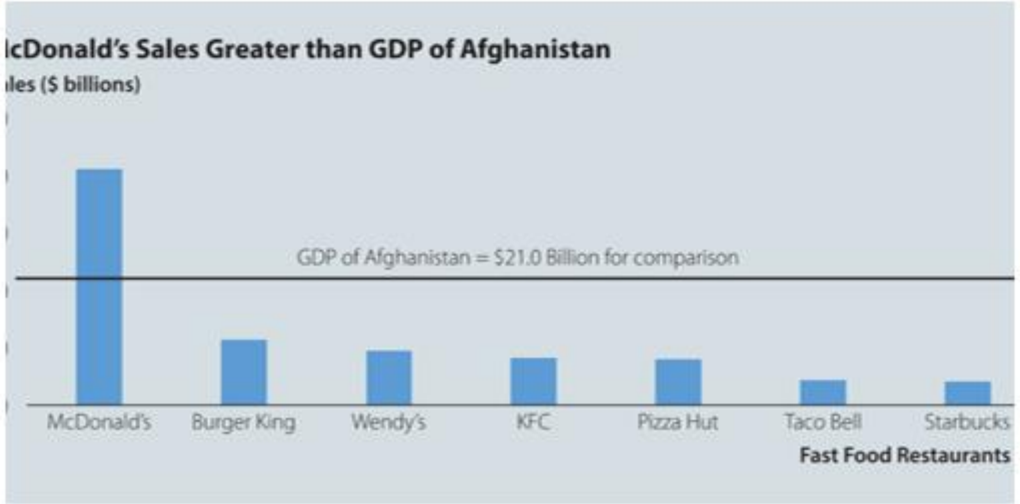
Data Visualization Makeover

Column chart using logos to compare fast-food restaurant sales



(Source: http://www.princeton.edu/~ina/infographics/starbucks.html)

Improved column chart to compare fast-food restaurant sales



Visual Perception and Preattentive Attributes

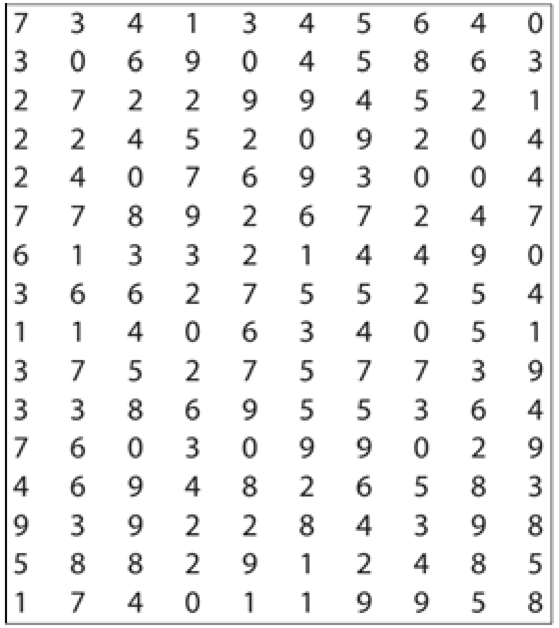
- **Visual perception** – brain process that interprets reflections of light that enters eyes
- Three forms of memory affect visual perception:
    - **Iconic memory**
    - **Short-term memory**
    - **Long-term memory**
- **Preattentive attributes** – features processed by iconic memory.

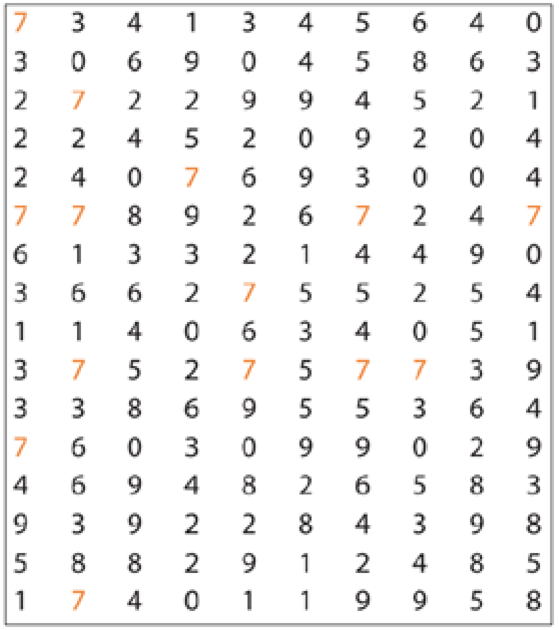
Cognitive Load and Preattentive Attributes

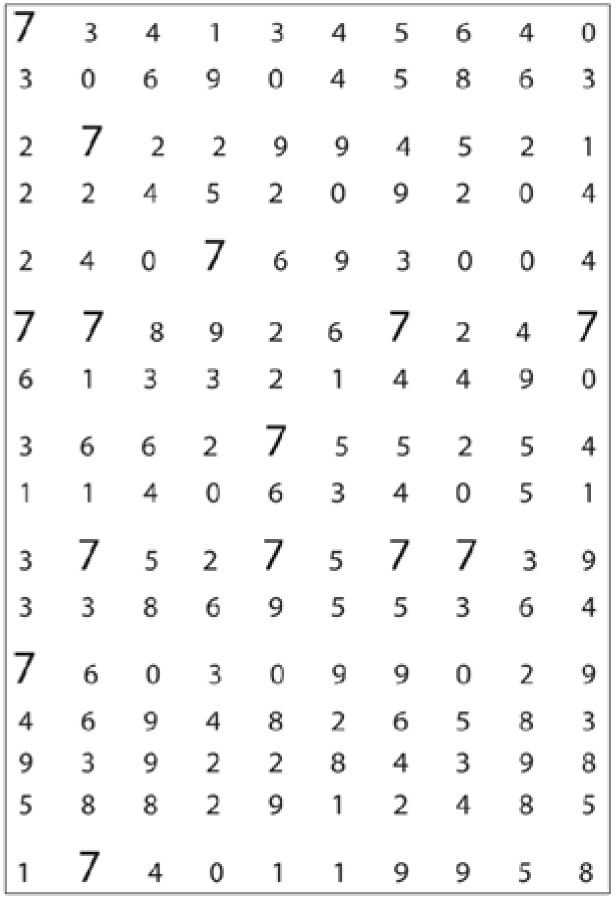
- **Cognitive load** – the effort needed to accurately and efficiently process information communicated by a data visualization
  - Proper use of preattentive attributes reduces the cognitive load
- Four preattentive attributes related to visual perception are:
  - **Colour**
  - **Form**
  - **Spatial positioning**
  - **Movement**

Application of Preattentive Attributes

Count the number of 7s aided by the preattentive attributes of colour and form (size)

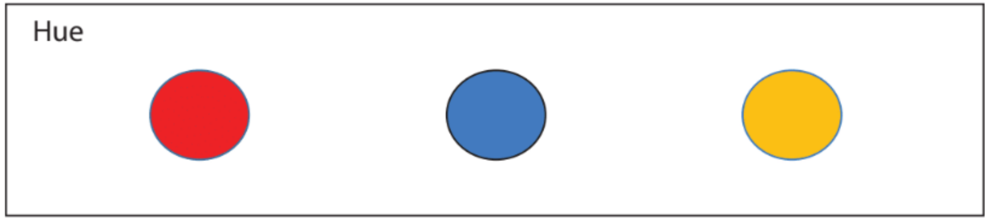


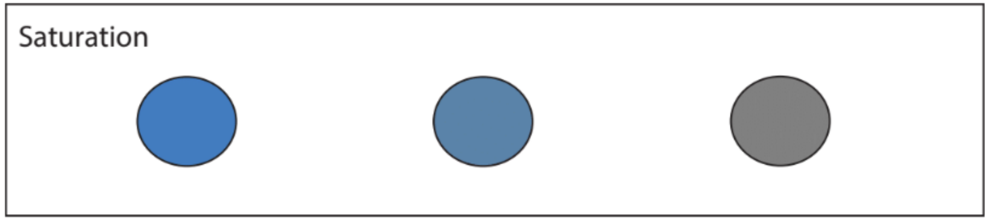


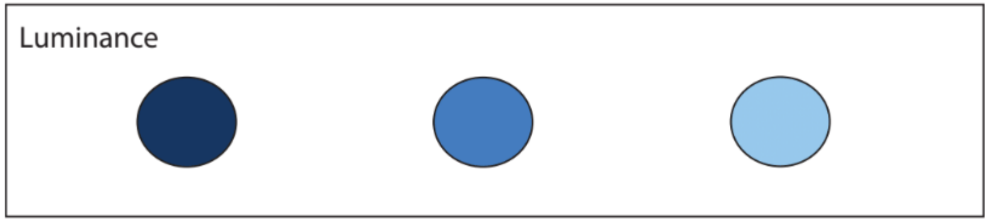


The Preattentive Attribute of Colour

The preattentive attribute of colour includes hue, saturation, and luminance



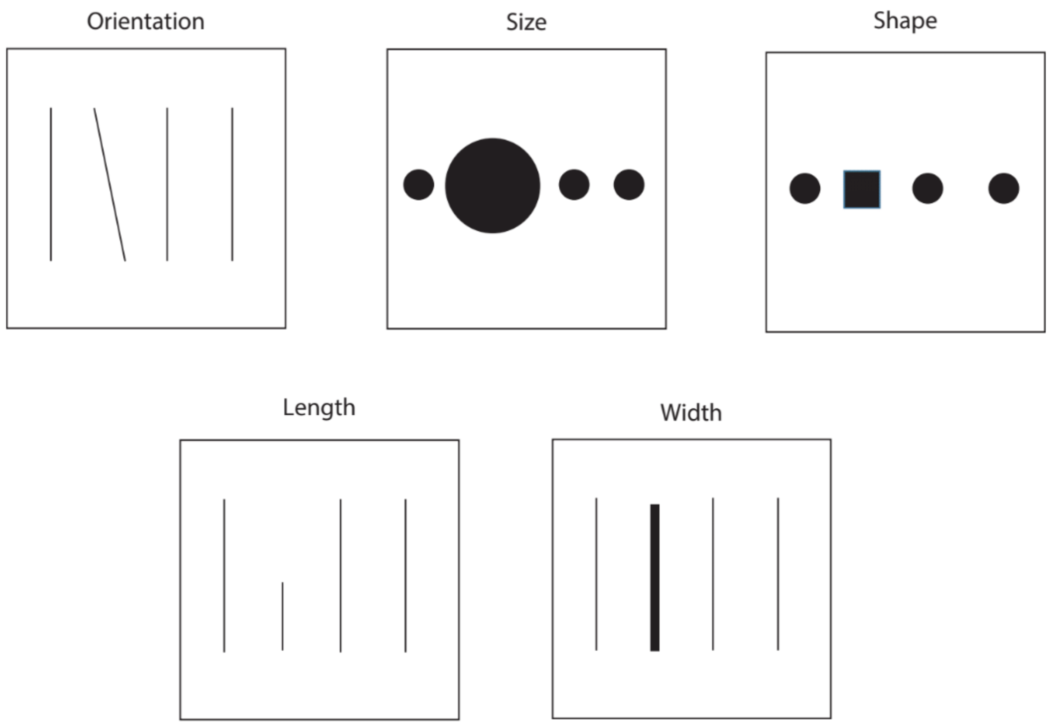




- **Hue** is the position light occupies on the visible light spectrum. It is the basis of the different colours.
- **Saturation** is the amount of gray present in colour. It refers to the intensity or purity of the colour.
- **Luminance** refers to the amount of black versus white within the colour.

The Preattentive Attributes of Form

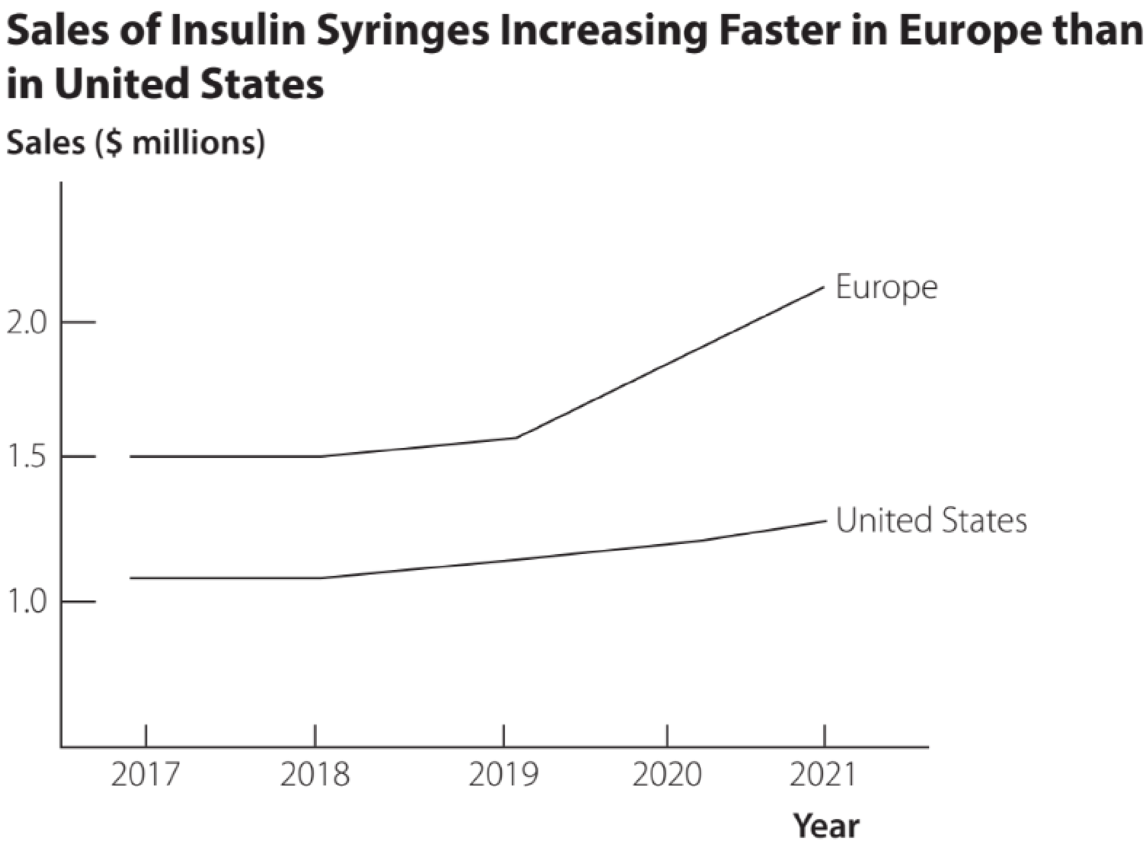
The preattentive attributes of form include orientation, size, shape, length, and width



- **Orientation** refers to the relative positioning of an object within a data visualization. It is common in line graphs.
- **Size** refers to the relative amount of space an object occupies in a visualization. Most people struggle to estimate relative size differences.
- **Shape** refers to the form of objects used in data visualization to distinguish between groups.
- **Length** and **Width** refer to the distance and thickness of a line or bar/column, respectively.

The Preattentive Attribute of Orientation

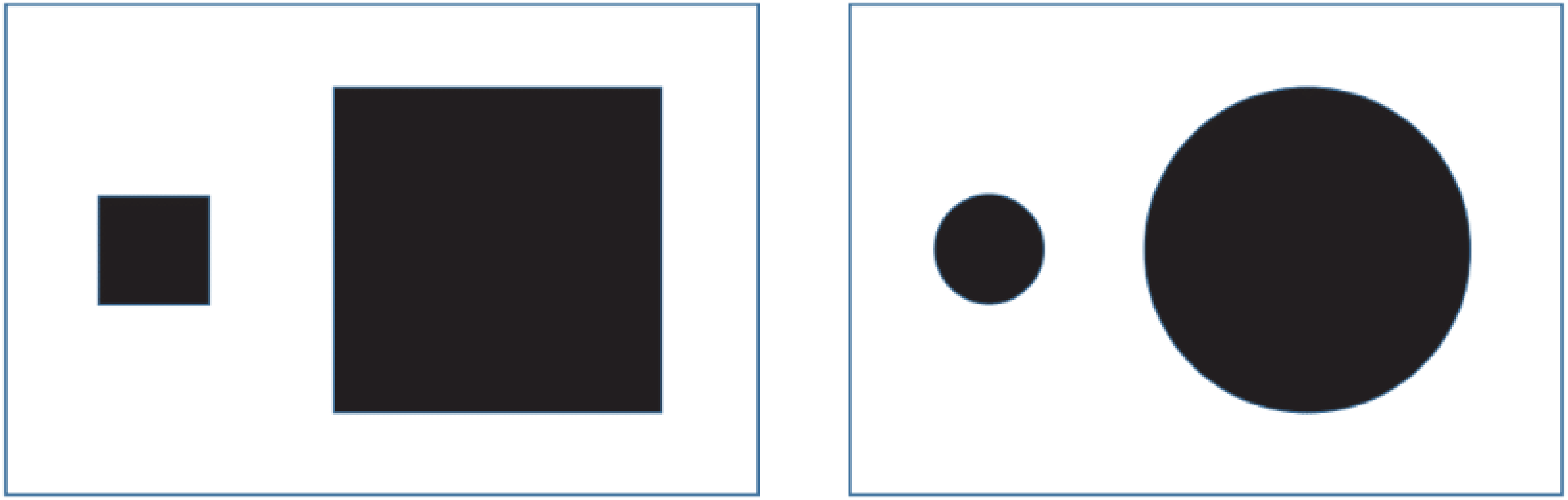
Demonstration of the preattentive attribute of orientation



The difference in the orientation of these lines makes it easy for the audience to perceive that sales in Europe are increasing at a much faster rate than the United States for the years 2019 and 2020.

The Preattentive Attribute of Size

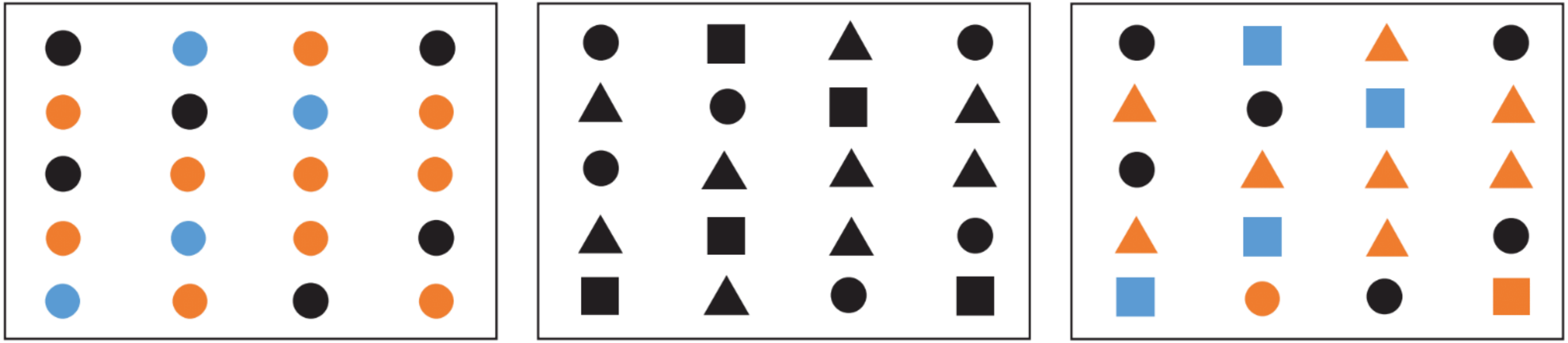
Demonstration of the preattentive attribute of size



Most people are not good at estimating relative size differences using the attribute of size.

The Preattentive Attribute of Shape

Grouping items by colour and shape

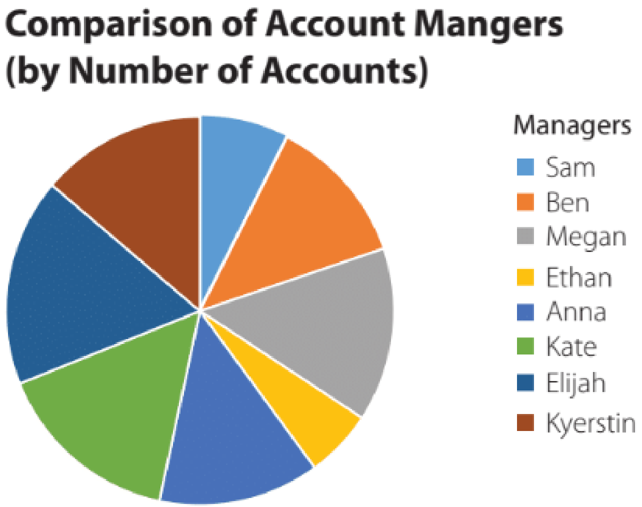


(a) (b) (c)

a. The preattentive attribute of colour divides the items into three different groups: orange, blue, and black.
b. The preattentive attribute of shape divides the items into three groups: circle, square, and triangle.
c. The combination of preattentive attributes of colour and shape divides the items into nine groups, each with one of the three colours combined with one of the three shapes

The Preattentive Attribute of Length

The bar chart (preattribute of length) makes it easier to display the accounts managed data than the pie chart (preattributes of size and colour)

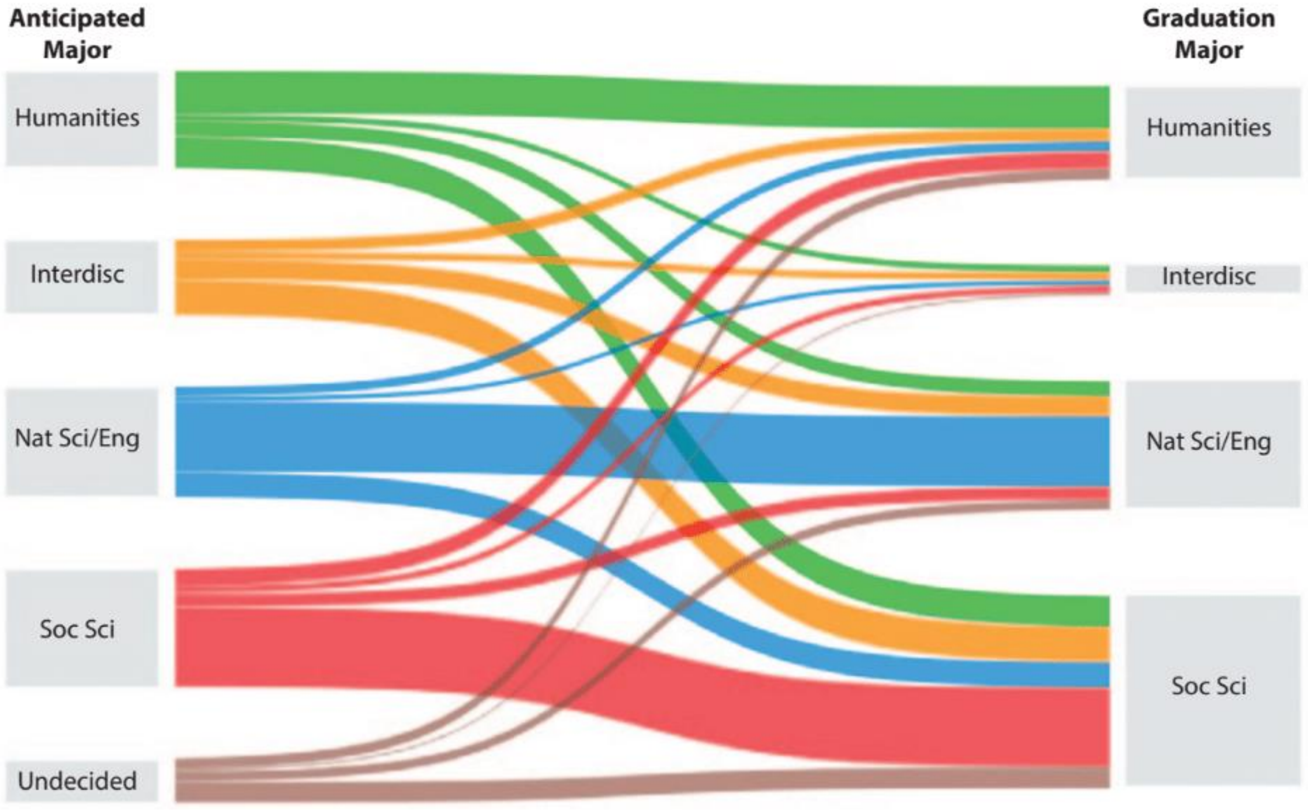


Line, bar, and column charts often employ the preattentive attributes of length and width:

- The **Length** refers to the horizontal, vertical, or diagonal distance of a line or bar/column.
- The **Width** refers to the thickness of the line or bar/column.

The Preattentive Attribute of Width

A Sankey chart demonstrating the use of the line width attribute



A **Sankey chart** typically depicts the proportional flow of entities where the width of the line represents the relative flow rate compared to the widths of the other lines.

In the example shown, the graduation major for students in a liberal arts college flows from the anticipated major.

The line width attribute is used much less frequently in data visualizations.

The Attributes of Spatial Positioning and Movement

- **Spatial positioning** refers to the location of an object within some defined space.
    - The most used spatial positioning in tables and charts is 2-D, such as the data points in a scatter chart.
- **Movement** can be categorized into two main attributes:
    - **Flicker** refers to effects such as flashing to draw attention.
    - **Motion** involves directed movement and can be used to show changes within a visualization.
- Movement is not possible in most data visualizations because tables and charts are static.

End of Part 1